Name: _____

Write your name and computing ID above. Write your computing ID at the top of each page in case pages get separated. Sign the honor pledge below.

On my honor as a student I have neither given nor received aid on this exam.

_____

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1. Suppose we have a channel on which we can transmit bits, and we want to use it to transmit distinct messages.

   Consider the following proposals:

   A. have messages be a fixed size and followed by a 32-bit checksum;
   B. have messages be fixed sized, with the bit pattern `0111111110` added before the message contents and a 32-bit checksum added after it;
   C. have messages be variable sized, and prefixed by their length (as a 32-bit number) and a 32-bit checksum;
   D. have messages be variable sized, and prefixed by the bit pattern `0111110` and have all `011111`s in the message body replaced with `0111111`

(a) Identify which of the protocols above have the following properties. (Write the letters of the protocols.)

   i. (2 points) Allows the bits for a message to start being sent before the entire contents o the message is known.

   AB

   ii. (2 points) Can usually avoid receiving corrupted messages when bits are flipped.

   ABC

   iii. (2 points) Can efficiently recover from bits being omitted or duplicated due to clock synchronization problems.

   BD

(b) (4 points) Suppose we are sending random messages of 2^{20} bytes (2^{23} bits). Order the proposals above by the expected total overhead (in terms of extra bits sent on the channel), with the highest-overhead proposal listed first:

   D > C > B > A

2. Suppose two hosts A and B are communicating via TCP over IPv6. They both have direct connections to a router C, for which they use a maximum segment size (maximum size of data in a TCP packet) of 1000 bytes. In the largest TCP packets they transmit, they have 1000 bytes of data, plus a 40-byte IPv6 heaer, plus a 32-byte TCP header (including 12 bytes of TCP options).

   Each Ethernet connection has a latency of 10 microseconds, a packet with the maximum amount of data takes 1 microsecond to transmit, a packet with no data takes around 0.1 microsceonds to transmit, and the router has an average queuing latency of 5 microseconds.

(a) (4 points) What is the approximate maximum transmission rate at which A can send data to B using their TCP connection? You may leave your answer as an unsimplified arihtmetic expression.

   1000 bytes / 1 microsceond

(b) (4 points) If we use a fixed window size for tranmission, then approximately what should that window size by in bytes to achieve the maximum tranmission rate? You may leave your answer as unsimplified arithmetic expression.

   1000 bytes / 1 microsecond * 2 * (1 + 10 + 5 + 1 + 10)

(c) (4 points) Suppose instead of having direct connections to C, they use a VPN-style tunnel to connect to C. This tunnel is implemented using GRE over IPv6, with a 40-byte outer IPv6 header specifying a destination of C, a 4-byte GRE header, followed by IPv6 and TCP headers in the original format.

   This reduces the maximum segment size to 996 bytes. What is the total increase in IPv6 packet bytes sent from A to B, in precent? Please leave your answer as an unsimplified arithmetic expression.

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Solution: old per packet = 1000+32+40; new per packet = 996+32+40+4+40; 1000/996 new packets per old packet, so 1000/996*(996+32+40+4+40)/(1000+32+40)*100 percent

3. Suppose three flows A, B and C go through a shared link, and the router that feeds that link uses fair queuing.

   The link can handle 1000 packets/second, flow A sends 500 packets/second, flow B sends 300 packets/second and flow C sends 300 packets/second, then

(a) (3 points) What is the expected packet loss rate (in percent) for flow A? You may leave your answer as an unsimplfied arithmetic expression.

Solution: 60

(b) (3 points) What is the expected packet loss rate (in percent) for flow C? You may leave your answer as an unsimplified arithmetic expression

Solution: 0

4. Suppose there is an active TCP connection between host A and host B and packets with the following attributes of packets sent from A to B:

from	to	sequence number	acknowledgment number	data size	dropped?
A	B	500	2310	1000	no
A	B	100	3310	0	no
A	B	1500	4310	0	no
A	B	1500	4310	1000	no
A	B	2500	4310	1000	no
A	B	3500	4310	1000	yes
A	B	4500	4310	1000	no
A	B	5500	4310	500	no

Assume that only packets labeled as 'dropped' are lost for any reason.

Recall that in TCP, the sequence number represents the **byte number** of the first data byte in that packet (so a 10 byte packet with sequence number 10 contains byte numbers 10 through 19, inclusive), and an acknowledgment number contains the byte number of the next expected sequence number (so acknowledgment number 20 acknowledges bytes up to and including 19).

(a) (4 points) Suppose B's sending window size was initially 2000 and it was not in 'slow start'. Assuming B most recently sent two 1000-byte data packets and was using the standard loss-based TCP congestion control with additive increase (by around 1 packet per round-trip-time) and multiplicative decrease (by a factor of two on loss or timeout), we would expect B's sending window size to _____ as result receiving A's messages above.

- ○ increase to around 2500
- ■ increase to around 3000
- ○ increase to around 4000
- ○ decrease to around 1000
- ○ something else, _____

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(b) (4 points) Suppose A sends the above packets without receiving packets from B between sending the first and last packets above. What is the **minimum** sending window size (in bytes) A could have?

5500

(c) (4 points) The second and third packets A sent were sent in response to packets B sent. What could have been attributes of those packets?

from	to	sequence number	acknowledgment number	data size
B	A	less than 331	500	???
B	A	331	500	100

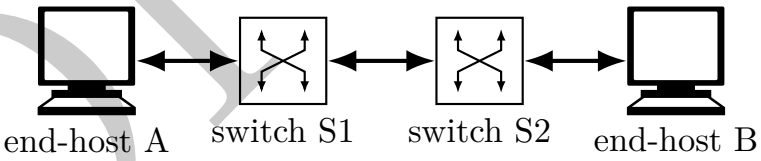
(d) (8 points) After these packets are sent, assuming B sends an acknowledgment packet for each packet it receives, complete the table of packets B will send to A:

from	to	sequence number	acknowledgment number	data size
B	A	4310	1500	0
B	A	4310	2500	0
B	A	4310	3500	0
B	A	4310	3500	0
B	A	4310	3500	0

(e) (4 points) After A receives the acknowledgment packets from B and any relevant timeouts expire, how will A change it sending window size and what, if any, data packet will A send next?

Solution: increase slightly for first two packets then halve on duplicate ACK; resend data with sequence number 3500 next

5. (6 points) Consider a network where machine A and machine B are connected via two switches S1 and S2:



A 10000-bit packet X from machine A, goes across a link to S1, then it goes across a link from S1 to S2, then it goes across a link from S2 to B.

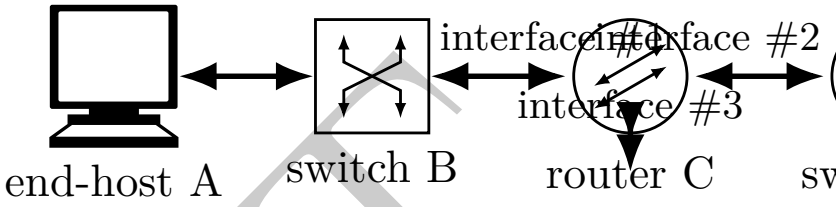
Each link has is composed of 100 meters of copper that has a bandwidth of 1Gbit/sec and a signal propogation rate of 2.8×10^8 m/s. Each switch, on average, has 2 other 10000-bit packets waiting to be transmitted when the packet X arrives.

What is the approximate latency of packet X from machine A to machine B? You may leave your answer as an unsimplified arithmetic expression.

*transmission time per link = 10000 ns; propogation per link = $\frac{100}{2.8 \times 10^8}$ s = 357 ns; queuing = 10000 ns * 2; total = (10000 + 357) * 2 + (10000 * 2) = 40714 ns*

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Solution: 40714



6. Suppose we have the following network topology:
(or described without a picture: end-host A and router interface 1 are connected to switch B; end-host E and F and router interface 2 are connected to switch; and router interface 3 is connected to the internet via the 203.0.113.0/24 network and the gateway 203.0.113.1).

   On these networks:

   - the end-host and router interfaces are assigned IP and MAC addresses as follows:

node/interface	IP	MAC
end-host A	`10.1.0.2`	`11:22:33:44:aa:bb`
end-host E	`10.2.0.2`	`11:22:33:44:aa:cc`
end-host F	`10.2.0.3`	`11:22:33:44:aa:cc`
router C interface 1	`10.1.0.1`	`11:22:33:44:11:22`
router C interface 2	`10.2.0.1`	`11:22:33:44:11:23`
router C interface 3	`203.0.113.5`	`11:22:33:44:11:24`

   - the router does **network address translation** for end-hosts A, E, and F using its interface 3 conenction
   - the router uses a gateway of 203.0.113.1 to contact hosts on the internet not in the 203.0.113.0/24 network

(a) (8 points) Suppose end-host A port 5555 sends a UDP packet to end-host E port 6666.

Fill in the table of the source and destination MAC addresses, IP addresses, and port numbers that will be present in those frames. For MAC addresses, please write only the last 16 bits (4 hexadecimal digits).

Assume that any ARP (address resolution protocol), DHCP, and related setup has already completed.

If you need to know an value that is not specified above, indicate something like 'X's MAC address'. If some field won't be present in the frame, write 'N/A'.

You may not need all lines.

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frame #		source	destination
1	last 16 bits of MAC	aa:bb	11:22
	IP address	10.1.0.2	10.2.0.2
	port numbers	5555	6666
2	last 16 bits of MAC	11:23	aa:cc
	IP address	10.1.0.2	10.2.0.2
	port numbers	5555	6666
3	last 16 bits of MAC		
	IP address		
	port numbers		
4	last 16 bits of MAC		
	IP address		
	port numbers		

(b) (5 points) Suppose end-host E makes a TCP connection from port 4444 to the internet host 192.0.2.10 port 443. Then it makes a TCP connection from port 4445 to internet host 192.0.2.11 port 443. The end-host successful completes these TCP connections, but does not send data over them.

When this happens, _____. **Select all that apply.**

- ■ at least two packets are received by end-host E
- ☐ end-host E will receive information sufficient to determine that the router's internet IP address is 203.0.113.5
- ■ the IP packets received by 192.0.2.10 will have a different checksum in the IP header than the corresponding IP packets that end-host E sent
- ☐ the TCP packets sent by the router over its internet connection will use consecutive source port numbers
- ☐ 192.0.2.10 will receive information sufficient to determine that end-host E's MAC address is 11:22:33:44:aa:cc

(c) (8 points) Complete a likely routing table for router C. Use CIDR notation (e.g. 1.2.0.0/16) for all address ranges. You may not need all lines.

destination IPs	gateway (if applicable)	interface #
10.1.0.0/16 or similar	N/A	1
10.2.0.0/16 or similar	N/A	2
203.0.113.0/24	N/A	3
0/0 or default	203.0.113.1	3
—	—	—
—	—	—
—	—	—

7. Consider the following network topology:
   - router A is connected to:
     - network 1
     - network 2

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  - network 3
- router B is connected to:
  - network 1
  - network 4
  - network 5
- router C is connected to:
  - network 5
  - network 6
- router D is connected to:
  - network 4
  - network 6
- router E is connected to:
  - network 6
  - network 7
- router F is connected to:
  - network 2
  - network 4

(a) Suppose the routes were setup via a distance vector protocol, where each router is considered to have a cost of 1 to reach the networks it is connected to and any router attached to any of those networks. Assume the 'split horizon with poison inverse' approach is in use, where distance vectors advertized to neighbors have a cost of infinity for routes that go through that neighbor. (For example, if D's route to network 7 uses router E, it will advertise a distance vector with cost ∞ for network 7 to router E, but one with a finite cost to router A.)

i. (5 points) Suppose the routes are setup using the distance vector protocol and then router D goes offline. After the routers that are connected to the same networks as it detect this, idenitfy which routers will update their distance vectors as a direct or indirect result of router D going offline. **Select all that apply.**

- ☐ A
- ☐ B
- ☐ C
- ☐ E
- ☐ F

ii. (4 points) If router E's link to network 7 went offline (but other network connections remained active), then the count-to-infinity problem could result in an indefinite change of distance vector updates. While this happens, which of the following might be true? **Select all that apply.**

- ■ the route from router A to network 7 could go in a loop through routers B, C, F and then back to A (without going through other routers)
- ☐ the route from router A to network 7 could go in a loop through router B and then back to router A (without going through other routers)
- ☐ the route from router A to network 6 could go in a loop through routers B and F and then back to A (without going through other routers)
- ☐ there would be no working route from network 2 to network 6

(b) (6 points) Rather than decide routes using a distance-vector or link-state routing protocol, we could decide which connections from one router to another to enable using a spannning tree protocol.

Suppose we compare the results of a spanning-tree protocol (with some assignment of ID numbers to the routers) to using a link-state algorithm like OSPF where each router is considered to have a cost of 1 to reach the networks it is connected to and any router attached to any of those networks.

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Give an example of a set of paths the OSPF-like protocol could select that the spanning tree protocol would not, even if we tweaked the spanning tree protocol by selecting different ordering for the routers (for example, to select a particular router as the root of the spanning tree).

Solution: example: router A to D via F (and network 2+4); router D to C directly (via network 6); router C to A via B (and network 5+1)

8. Suppose we have:

- a router that:
  - is connected to a local 10.0.0.0/24 Ethernet network with IP address 10.0.0.1
  - is connected the Internet with IP address 198.51.100.45 and gateway 198.51.100.1
  - uses network address translation to allow machines on the 10.0.0.0/24 network to access theInternet using the router's Internet IP address
- a machine with a web browser in the local Ethernet network with IP address 10.0.0.4
- a machine with a web proxy in the local Ethernet netiwork with IP address 10.0.0.7

The web browser on 10.0.0.4 makes the following request to the web proxy:

```
GET http://quux.foo.com/ HTTP/1.1
Host: quux.foo.com
Cookie: foo=bar
```

(a) (8 points) Suppose the web proxy looks up the IPv4 address of `quux.foo.com`. Rather than contacting a recursive DNS resolver, it makes DNS queries itself starting with a root server via IPv4. It sends three separate queries. Complete the table with plausible information about these queries and their response. (You may not need all lines.) *one possible answer:*

destination hostname	query question	response
a.root-servers.net	quux.foo.com A	com. NS a.gtld-server.net. a.gtld-servers.net. IN A 192.5.6.30
a.gtld-server.net	quux.foo.com A	foo.com NS ns.bar.com ns.foo.com IN A 1.2.3.4
ns.bar.com	quux.foo.com A	quux.foo.com IN A 2.3.4.5

(b) (4 points) The web proxy sends a DNS query to a.root-servers.net (which has IPv4 address 198.41.0.4) via UDP over IPv4. Before it is processed by routers, the DNS packet the web proxy sends for this query will have a destination IP address of 198.41.0.4. When a.root-servers.net sends its response to the query, before any processing by routers, the DNS packet it sends will have a destination address of 198.51.100.45.

If not enough information is provided to answer above, explain briefly below.

Solution:

(c) (4 points) After resolving the IP address of quux.foo.com, the web proxy uses the sockets API to send a request to it. When it does so, it is likely to _____. **Select all that apply.**

- ☐ pass the local IP address of the router (10.0.0.1) to the 'connect' call for that socket
- ■ pass the default port number the HTTP protocol (80) to the 'connect' call for that socket
- ☐ pass the local IP address of the router (10.0.0.1) to the 'bind' call for that socket.

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☐ call the listen function on that socket, so it can use it as a server socket and accept connections

(d) (4 points) Suppose the web proxy makes a request to the quux.foo.com server, then gets the reply:

```
HTTP/1.1 200 OK
Cache-Control: max-age=600
Content-Type: text/plain
Content-Length: 10000
E-Tag: "abcd-12345"

...
```

(where ... represents some unshown bytes).

Which of the following might the web proxy do with this reply? **Select all that apply.**

- ■ send it to 10.0.0.4 again when it makes another request for the same URL immediately afterwards, without sending another request to quux.foo.com
- ■ send it to 10.0.0.4 again when it makes another request for the same URL later after recieving a "304 Not Modified" response from quux.foo.com
- ☐ send it to 10.0.0.4 again when it makes a request for the URL http://www.quux.foo.com/ without sending another request ot quux.foo.com or www.quux.foo.com
- ■ send it to another web browser (for example, on 10.0.0.3) that makes a request shortly after 10.0.04's request that is identical, except it has a different value for the Cookie header, wihtout sending another request to quux.foo.com

9. Suppose there are four nodes A, B, C, D, and E using a Wi-Fi-style wireless network to communicate. Nodes are in radio range to reach other as indicated in the following table:

	A	B	C	D	E
A	✓	✓	✓		✓
B	✓	✓		✓	✓
C	✓		✓	✓	✓
D		✓	✓	✓	✓
E	✓	✓		✓	✓

Assume that

- if a node receives tranmissions from two nodes at the same time, the result is a collision so neither transmission is intelligible;
- a node cannot receive other transmissions while it is also transmitting
- that propogation delays are very small compared to message transmission times or the length of time used for carrier sense.

(a) (4 points) When they detect another transmission with carrier sense, sometimes wireless protocols wait a random amount of time after the current transmission is finished before attempting a new transmission.

In which scenarios would this strategy be likely to improve performance?

- ■ B and C are both transmitting messages to A
- ■ B and C and E are all transmitting messages to A
- ○ D is transmitting messages to C and B is transmitting messages to A
- ■ E is transmitting messages to B and A is transmitting messages to C

(b) (4 points) Suppose we are using carrier sense with randomized exponential backoff and no ready-to-send/clear-to-send protocol or similar.

Also suppose that if A was transmitting a large stream of data to B, then the effective bandwidth of that stream is almost the capacity of the wireless channel.

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If, instead, we have A transmitting a large stream of data to B **and** E transmititng a large stream of data to C **and** B transmitting a large stream of data to E at the same time. Assume they always have packets queued to transmit. The combined effective bandwidth (after accounting for retransmissions, lost packets, etc.) of these three streams will be _____.

- ○ very small — almost all packets will not be transmitted successfully
- ■ much less than the total bandwidth with one sender, but still significant
- ○ measurably less than the total bandwidth with one sender, but almost all the channel's capacity
- ○ essentially the same as the total bandwidth with one sender
- ○ more than the capacity of the wireless channel (more bandwidth because of three separate streams)

(c) (4 points) For which of the following scenarios would ready-to-send/clear-to-send improve performance? **Select all that apply.**

- ■ A is transmitting a large stream of data to B while D is transmitting a large stream of data to B
- ■ A is transmitting a large stream of data to B while D is transmitting a large stream of data to C
- ☐ B is transmitting a large stream of data to A while D is transmitting a large stream of data to C
- ☐ A is transmitting two large streams of data, one to B and one to C

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